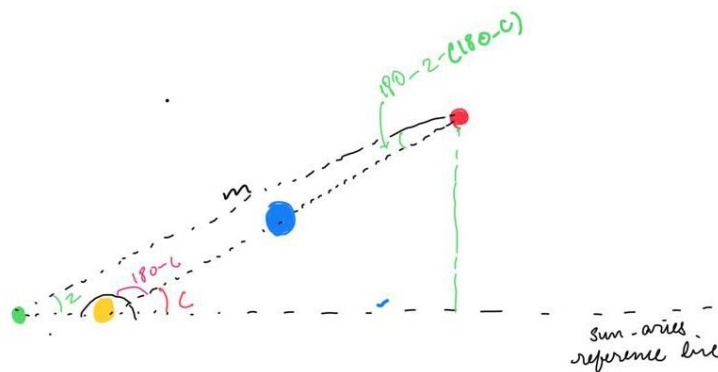
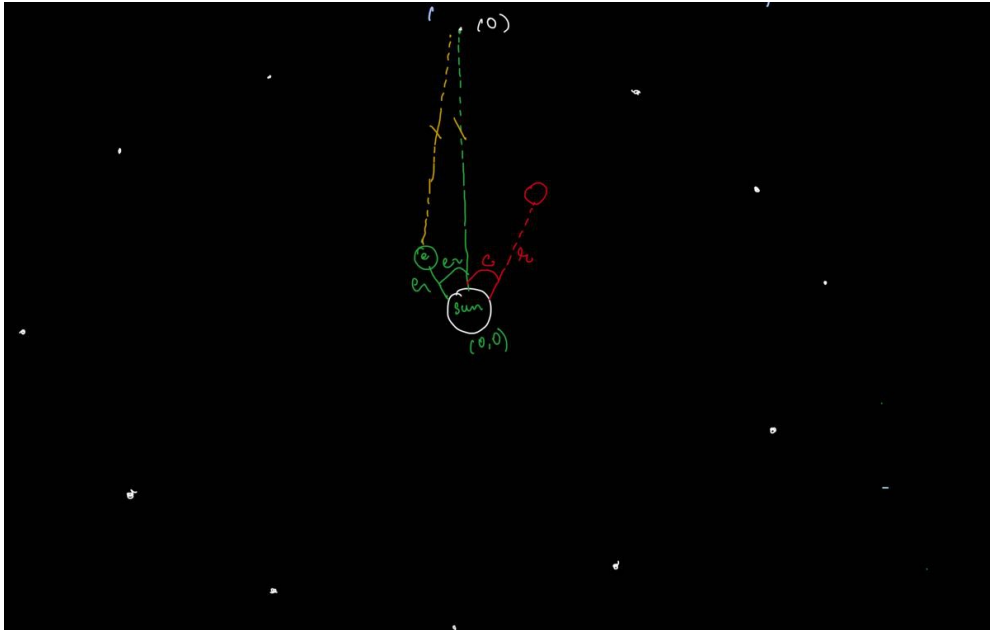


Data Analytics E0-259

Assignment 2

Yogangi Tiwari

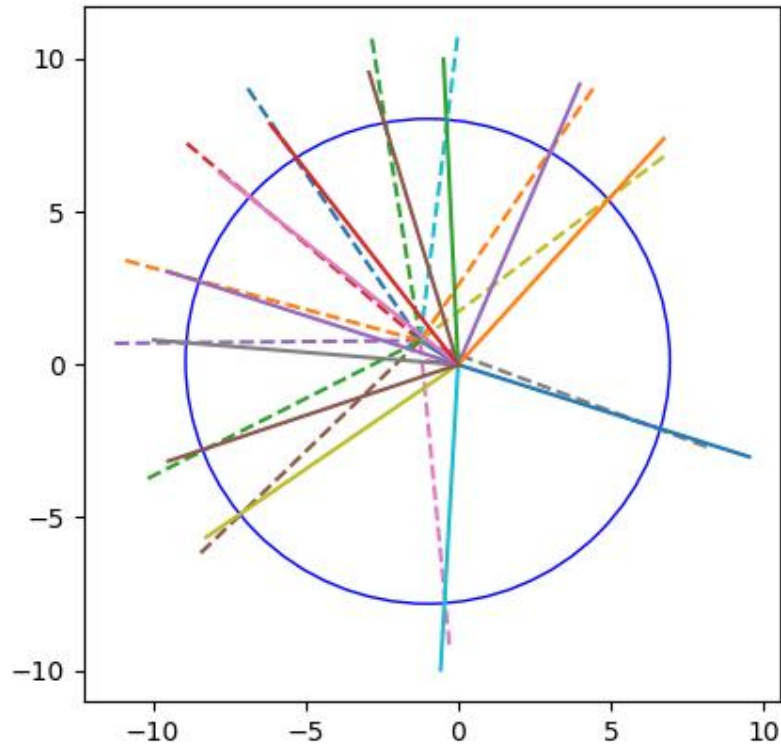
SR. No. 24146



Output

Parameters/ Arguments	Value	Unit
E1	1.4707930986350832	AU
E2	2.575078146988676	radians
S	0.009147067909857911	radians
R	7.929643498586542	AU
c	3.0303151267278743	radians

Max error = 0.0297 i.e. 1.7825 degrees.



Implementation Summary

- To determine Mars' orbit, its position needs to be determined
- Objective is attained by taking a hypothetical intersection from an imaginary point (equant) and heliocentric longitudes.
- Equant is a position such that mars' angular velocity around it is considered constant.
- To obtain intersection points following is the calculation:

A circle drawn from mars orbit centre; the eq. is as follows:

$$(x - c)^2 + (y - c)^2 = r^2$$

A line passing through equant centre and making z' angle from Aries reference line:

$$(y - ey) = (x - ex) \tan(z')$$

$$z' = (z + s \cdot T)$$

Solving above equations will result in (x_1, y_1) and (x_2, y_2) :

The intersection point (circle and line) near to opposition is selected to calculate the errors.

Question 1

- The primary objective is to iteratively optimize a set of orbital parameters for Mars and minimize the angular errors between the observed and predicted opposition positions.
- Errors and max errors were obtained by first running an exhaustive search for e_1 , e_2 , c and z , given R and S , started with initial guess of c , e_1 , e_2 , z , along with the calculated errors and the maximum angular error.
- Mars equant model was initiated to obtain the required arguments. The looped optimization process ensures a thorough search across the parameter space.
- `run_exhaustive` is a helper function that optimizes the parameters c , e_1 , e_2 , and z using a sequence of single-value optimizations.
- For each parameter, it runs `SingleOptimizer`, which adjusts that parameter based on the Mars model and the given bounds.
- After optimizing each parameter individually, the `InternalOptimizer` is called to further fine-tune all four parameters together.
- After all iterations, the model is fitted to calculate the final maximum error and opposition errors using the `fit` and `measured_err` methods of `MarsModel`.
- It returns the optimized values of c , e_1 , e_2 , and z , along with the calculated errors and the maximum angular error.

Question 2

- First running an exhaustive search for e_1 , e_2 , c and z , given R and S , started with initial guess of c , e_1 , e_2 , z .
- A dictionary `q2_args` is created to store the initial guesses.
- The function `bestOrbitInnerParams` is called to compute the best values of the parameters based on the provided guesses (R , S , and `q2_args`).
- The `times` and `oppositions` arrays (`time_series` and `longitude_series`) represent the opposition data for Mars and is further used to compute the angular errors.
- The function `bestOrbitInnerParams` performs iterative optimization, refining the parameters c , e_1 , e_2 , and z while keeping $r=R$ and $s=S$ fixed. It searches for the best fit that minimizes the angular error between the observed and predicted oppositions.

Question 3

- A discretised search for s with parameter r , `times` and `oppositions`.
- Calls the `bestOrbitInnerParams` function, which computes the orbital parameters c , e_1 , e_2 , z , and the angular errors (`errors` and `max Error`).
- The `objective_3` function is the objective function to be minimized during the optimization process. It takes the parameter s .
- `bestS` is responsible for optimizing s by minimizing the output of `objective_3`, which returns the maximum angular error.
- Once s is optimized, the `MarsEquantModel` is called using the updated parameters (c , r , e_1 , e_2 , z , and s) to compute the angular errors between the predicted and observed oppositions of Mars and returns optimised s .

Question 4

- Optimized S from above other parameter is used as initialization. The R is calculated. The function uses the same concept as the above function `bestOrbitInnerParams`.
- `Objective_4` is defined to take r as input and calculate the maximum angular error using `bestOrbitInnerParams`.
- `Minimize` is used to find the optimal r, method used is Nelder-Mead with a tolerance ($\text{tol}=1\text{e-}16$) to achieve a highly precise result.

Question 5

- Wrapper function optimizes all the parameters as a whole and gives best parameter by optimizing all the values together.
- The goal of this function is to find the optimal values of the parameters r, s, c, e1, e2, and z that minimize the maximum angular error between the predicted and observed positions of Mars (oppositions) at specific times.
- The in-built `minimize` function from `scipy.optimize` is used to find the optimal values for r and s using the Nelder-Mead method as it is well-suited for non-linear optimization.
- The `objective_rs` function is minimized, and the result returns the optimized values for r and s.
- `bestOrbitInnerParams` function is called again to optimize the other parameters (c, e1, e2, and z) based on the new values of r and s.